AN ETALE MORPHISM PRESERVES THE DIMENSION OF A SCHEME AT A POINT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Dimension at a point as the minimum of the Krull dimensions of the open neighbourhoods. Topology, Definition topology-definition-Krull, invoked at proof lines 5396-5399. Fixes the meaning of both sides of the asserted equality.
- The fibres of an etale morphism over a field are discrete. Morphisms, Lemma morphisms-lemma-etale-over-field, invoked at proof lines 5412-5414. Guarantees that the images of two distinct members of a chain of specializations in U stay distinct in V.
- Etale morphisms are open and flat, and a flat local ring map satisfies going down. Morphisms, Lemma morphisms-lemma-fppf-open at proof lines 5421-5422, Morphisms, Lemma morphisms-lemma-etale-flat at proof lines 5433-5434, and Algebra, Section algebra-section-going-up at proof lines 5437-5438. Openness keeps the image of a neighbourhood open, flatness makes the map of local rings flat and local, and going down lifts a chain of specializations along it.
- Points of a scheme and specializations as primes of a local ring. Schemes, Lemma schemes-lemma-specialize-points, invoked at proof lines 5435-5436. Converts the chain of points in V into primes of the local ring, which is what going down applies to.

Source reference. Original section 00HU: sources/stacks-chow/tags.txt, source lines 656–656. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 02GS: sources/stacks-chow/tags.txt, source lines 3239–3239. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02GL: sources/stacks-chow/tags.txt, source lines 3232–3232. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01J7: sources/stacks-chow/tags.txt, source lines 2004–2004. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 0055: sources/stacks-chow/tags.txt, source lines 197–197. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : U \to V$ be an étale morphism of schemes. Let $u \in U$ and $v = f(u)$. Then $\dim_u(U) = \dim_v(V)$.*

Proof. In the statement $\dim_u(U)$ is the dimension of U at u as defined in Topology, Definition 0055 as the minimum of the Krull dimensions of open neighbourhoods of u in U. Similarly for $\dim_v(V)$.

Let us show that $\dim_v(V) \geq \dim_u(U)$. Let V' be an open neighbourhood of v in V. Then there exists an open neighbourhood U' of u in U contained in $f^{-1}(V')$ such that $\dim_u(U) = \dim(U')$. Suppose that $Z_0 \subset Z_1 \subset \ldots \subset Z_n$ is a chain of irreducible closed subschemes of U'. If $\xi_i \in Z_i$ is the generic point then we have specializations $\xi_n \rightsquigarrow \xi_{n-1} \rightsquigarrow \ldots \rightsquigarrow \xi_0$. This gives specializations $f(\xi_n) \rightsquigarrow f(\xi_{n-1}) \rightsquigarrow \ldots \rightsquigarrow f(\xi_0)$ in V'. Note that $f(\xi_j) \neq f(\xi_i)$ if $i \neq j$ as the fibres of f are discrete (see Morphisms, Lemma 02GL). Hence we see that $\dim(V') \geq n$. The inequality $\dim_v(V) \geq \dim_u(U)$ follows formally.

Let us show that $\dim_u(U) \geq \dim_v(V)$. Let U' be an open neighbourhood of u in U. Note that $V' = f(U')$ is an open neighbourhood of v by Morphisms, Lemma 01UA. Hence $\dim(V') \geq \dim_v(V)$. Pick a chain $Z_0 \subset Z_1 \subset \ldots \subset Z_n$ of irreducible closed subschemes of V'. Let $\xi_i \in Z_i$ be the generic point, so we have specializations $\xi_n \rightsquigarrow \xi_{n-1} \rightsquigarrow \ldots \rightsquigarrow \xi_0$. Since $\xi_0 \in f(U')$ we can find a point $\eta_0 \in U'$ with $f(\eta_0) = \xi_0$. Consider the map of local rings

$$\mathcal{O}_{V',\xi_0} \longrightarrow \mathcal{O}_{U',\eta_0}$$

which is a flat local ring map by Morphisms, Lemma 02GS. Note that the points ξ_i correspond to primes of the ring on the left by Schemes, Lemma 01J7. Hence by going down (see Algebra, Section 00HU) for the displayed ring map we can find a sequence of specializations $\eta_n \rightsquigarrow \eta_{n-1} \rightsquigarrow \ldots \rightsquigarrow \eta_0$ in U' mapping to the sequence $\xi_n \rightsquigarrow \xi_{n-1} \rightsquigarrow \ldots \rightsquigarrow \xi_0$ under f. This implies that $\dim_u(U) \geq \dim_v(V)$. □